MODELLING AND ANALYSIS OF SOFT BENDING ACTUATOR

A report submitted in partial fulfilment of the requirements
for the award of the degree of

Bachelor of Technology

in

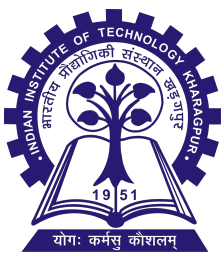
Mechanical Engineering

by

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November 2026

DECLARATION

I certify that

(a) the work contained in this report is original and has been done by me under the guidance of my supervisor;

(b) the work has not been submitted to any other institute for any degree or diploma;

(c) I have followed the guidelines provided by the Institute in preparing the report;

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CERTIFICATE

This is to certify that the report entitled **"Modelling and Analysis of Soft Bending Actuator"**, submitted by **Rudranarayan Pradhan** (Roll No. 23ME31022) to the Indian Institute of Technology Kharagpur in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Mechanical Engineering, is a record of the work carried out by the candidate under my supervision during the seventh semester (Autumn Semester, 2026–27).

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ABSTRACT

A soft pneumatic bending actuator is studied in which a silicone cylinder carries an off-centre (eccentric) cavity. When the cavity is pressurised the cylinder bends toward its thicker wall. The aim of the work is the relation between the cavity pressure and the bending of the actuator.

The cross-section properties, the eccentric thrust moment and the Euler–Bernoulli bending response are derived in closed form. A free-body argument shows that the thrust moment is uniform along the actuator for any material law. An equilibrium argument for the in-plane wall stresses then shows that, for a bare (unreinforced) actuator, linear elasticity multiplies the beam curvature by $(1 - 2\nu)$, which vanishes for nearly incompressible silicone. The bending that a bare actuator does show is therefore a finite-strain effect: the thin wall lengthens more than the thick wall, at second order in its hoop strain, and the response ends at a ballooning limit pressure.

These results were checked with nonlinear three-dimensional finite-element (FE) models built in COMSOL Multiphysics 6.4: hyperelastic (neo-Hookean) silicone with a nearly incompressible mixed formulation, a follower pressure on every face of the cavity, and an optional inextensible hoop sleeve modelled as a weak-form surface energy. For a placeholder design ($R = 10$, $r = 4$, $r_p = 3\,\mathrm{mm}$, $l = 100\,\mathrm{mm}$, $E = 0.6\,\mathrm{MPa}$) the tip angle of the sleeved actuator agrees with the beam formula within 1 % at $P/E = 0.017$ and within 5 % at $P/E = 0.033$, and exceeds it at higher pressure. The bare actuator bends only 11 % of the beam prediction at $P/E = 0.017$, and its response steepens strongly with pressure, as the analysis predicts.

A tapered actuator with four chambers, intended to bend in any direction, was then modelled in COMSOL with the real cavity pressure and nearly incompressible silicone. With an assumed modulus $E = 0.07\,\mathrm{MPa}$, the bare four-chamber actuator balloons at 11.7 kPa ($P/\mu = 0.50$), while the sleeved actuator bends smoothly to 47° at 10 kPa and 95° at 15 kPa. Beyond about 15 kPa the computed local stress and element distortion grow rapidly; whether this is physical or numerical is not yet resolved, so larger bending angles, contact and grasping forces are left for future work. All geometry and material values are placeholders; no experiment has been carried out yet.

Keywords: soft robotics, pneumatic actuator, hyperelasticity, ballooning, finite-element analysis, fibre reinforcement.

NOMENCLATURE

Symbol	Meaning	Unit
R	outer radius of the actuator	mm
r	cavity radius (single-cavity actuator)	mm
r_p	cavity eccentricity (offset of the cavity centre from the axis)	mm
l, L	pressurised length; total length of the four-chamber actuator	mm
$t_{\min}$, $t_{\max}$	thinnest and thickest wall, $R \mp r_p - r$	mm
$t(\phi)$	wall thickness along a ray at angle ϕ from the thin side	mm
A, A_c	elastomer area of the section; cavity area	mm^2
$\bar{y}$	position of the neutral axis	mm
I_{zz}	second moment of area about the neutral axis	mm^4
e	moment arm of the thrust about the neutral axis	mm
P	gauge pressure in the cavity	kPa
F, M	axial thrust on the tip cap; bending moment	N, N mm
κ, ρ	curvature; radius of curvature	mm^{-1}, mm
θ, δ	tip angle (tip rotation); transverse tip deflection	deg, mm
E, μ, ν	small-strain Young's modulus, shear modulus, Poisson's ratio	MPa, MPa, –
κ_b	bulk modulus	MPa
C_{10}, C_{20}, C_{30}	Yeoh coefficients (neo-Hookean: $C_{10} = \mu/2$, others zero)	MPa
λ, λ_h	stretch; hoop stretch	–
J	local volume ratio $\det \boldsymbol{F}$	–
$\boldsymbol{F}$	deformation gradient	–
k_s	hoop-sleeve stiffness (penalty)	N/mm
e_h	Green strain of the outer surface in the hoop direction	–
W	strain-energy density	MPa
$\bar{I}_1$	first invariant of the isochoric right Cauchy–Green tensor	–
ε_0, ε_x	centroidal axial strain; axial strain	–
σ_x, σ_y, σ_z	normal stresses	MPa
α	half-angle of a chamber sector (four-chamber actuator)	deg
$a(x)$, $b(x)$	inner and outer radius of a chamber at position x	mm
FE	finite element(s)	
NA	neutral axis	

CHAPTER 1

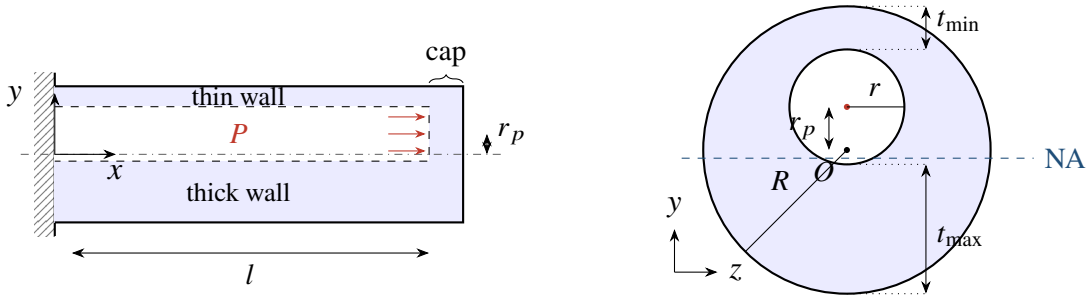
INTRODUCTION

1.1 Background

Soft robots built from elastomers can bend, grip and conform to objects without rigid joints, which makes them attractive for handling delicate or irregular objects and for medical devices. Among their actuators, fluid-driven elastomer actuators are the most common: a cavity in the elastomer is pressurised with air or liquid and the body deforms [1, 2]. How a given pressure turns into a given motion depends on the geometry, on the strongly nonlinear behaviour of the elastomer and on any reinforcement such as fibres or inextensible layers [3, 4]. A model of this relation is needed both to design the actuator and to control it.

1.2 The actuator studied

The actuator considered first is a solid silicone cylinder of outer radius R and pressurised length l, with a circular cavity of radius r whose centre is offset by r_p from the axis (Figure 1.1). One wall is therefore thin, $t_{\min} = R - r_p - r$, and the opposite wall thick, $t_{\max} = R + r_p - r$. The base is clamped, the far end is closed by a solid cap, and the cavity is pressurised to a gauge pressure P. The actuator bends toward the thick wall. The study is later extended to a tapered actuator with four chambers, which can bend in any direction by pressurising one chamber or a combination of chambers.



(a) Side view (bending plane x–y). **(b)** Cross-section.

Figure 1.1. Eccentric-cavity actuator: geometry and coordinate frame. The origin O is the centre of the outer circle, the cavity centre is at $(y, z) = (r_p, 0)$ and the neutral axis (NA) lies at $y = \bar{y} < 0$. Not to scale.

1.3 Objectives

1. Derive the pressure-to-bending relation of the eccentric-cavity actuator in closed form and state the assumptions under which it holds.
2. Build nonlinear three-dimensional FE models in COMSOL Multiphysics that do not rely on those assumptions, and use them to test the closed-form result for a bare and for a fibre-reinforced actuator.
3. Extend the model to a tapered four-chamber actuator for multi-directional bending and find how far it can bend under real pressure.

1.4 Scope and limitations

This report covers modelling only. The geometry and material values used are *placeholders* chosen to make the models concrete; the real design, the silicone and its measured constants are still to be fixed, and no experiment has been carried out. The silicone is modelled as neo-Hookean. Gravity, viscoelasticity and the Mullins effect are not modelled. Contact with objects, and hence grasping forces, are not part of the results reported here (Section 5.4).

1.5 Organisation of the report

Chapter 2 derives the analytical model. Chapter 3 describes how the FE models were built, loaded, meshed and solved in COMSOL. Chapter 4 presents the results for the single-cavity actuator and for the four-chamber actuator. Chapter 5 discusses what the results mean and what limits them, and Chapter 6 gives the conclusions and the planned next steps.

CHAPTER 2

ANALYTICAL MODEL

Units throughout: mm, N, MPa (1 MPa = 1000 kPa = 1 N/mm^2). Axes: x along the actuator (base at $x = 0$); y in the section from the outer centre toward the cavity centre; z completes the right-handed frame. Bending is about z, toward $-y$.

2.1 Assumptions

(A1) The material is isotropic, homogeneous, rate-independent and (nearly) incompressible.

(A2) The pressure is uniform in the cavity; gravity and external loads are neglected; the base is clamped and the tip is free.

(A3) *Beam model only:* plane sections remain plane, the cross-section keeps its shape, and every longitudinal fibre is in uniaxial stress.

Assumption (A3) holds when a hoop reinforcement carries the circumferential load. Section 2.5 shows that it fails for a bare actuator.

2.2 Cross-section properties

The section properties follow from the composite-area method and the parallel-axis theorem [5]. The elastomer area is the outer disc minus the cavity:

$$A = \pi\,(R^2 - r^2). \tag{2.1}$$

Treating the cavity as a negative area centred at $y = r_p$, the neutral axis lies at

$$\bar{y} = \frac{\pi R^2 \cdot 0 - \pi r^2 r_p}{\pi R^2 - \pi r^2} = -\frac{r^2\, r_p}{R^2 - r^2}, \tag{2.2}$$

on the thick-wall side of the outer centre. By the parallel-axis theorem applied to each disc,

$$I_{zz} = \left[\frac{\pi R^4}{4} + \pi R^2 \bar{y}^{\,2}\right] - \left[\frac{\pi r^4}{4} + \pi r^2 (r_p - \bar{y})^2\right]. \tag{2.3}$$

With $r_p - \bar{y} = R^2 r_p/(R^2 - r^2)$ the two shift terms combine to $-\pi R^2 r^2 r_p^2/(R^2 - r^2)$, so

$$I_{zz} = \frac{\pi\,(R^4 - r^4)}{4} - \frac{\pi R^2 r^2 r_p^2}{R^2 - r^2}. \tag{2.4}$$

The first term is the concentric hollow cylinder; the second is the stiffness lost by moving the hole off-centre.

2.3 Pressure loading

The pressure acts on the inside face of the tip cap, whose projected area is the cavity area,[1] giving the thrust

$$F = P\,A_c = P\,\pi r^2. \tag{2.5}$$

Its resultant passes through the cavity centre $y = r_p$, at a distance

$$e = r_p - \bar{y} = \frac{R^2\,r_p}{R^2 - r^2} \tag{2.6}$$

from the neutral axis, so the bending moment is

$$M = F\,e = \frac{\pi P R^2 r^2 r_p}{R^2 - r^2}. \tag{2.7}$$

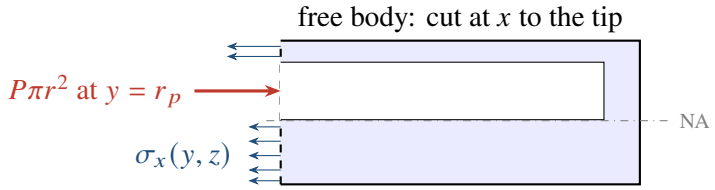


Figure 2.1. Free body from a cut at x to the tip, silicone together with the enclosed fluid.

The moment is uniform along the length. Cut the actuator at any x and take the free body from the cut to the tip *including the fluid it encloses* (Figure 2.1). No external force acts on it except at the cut, so

$$\int_A \sigma_x\,dA = P\pi r^2, \qquad \int_A \sigma_x\,(y - r_p)\,dA = 0. \tag{2.8}$$

The wall is in net tension $N = P\pi r^2$ acting through the cavity centre, which about the neutral axis is the moment $M = Ne$ of eq. (2.7), at every x and for any material law.[2] Equation (2.8) fixes only the *resultants* of σ_x; relating them to strain needs a constitutive assumption.

[1] The resultant of a uniform pressure on a surface, in a given direction, equals the pressure times the area of the surface projected on a plane normal to that direction; for the flat inner face of the cap this is simply the cavity area πr^2. The same argument gives the axial load in a closed thin-walled pressure vessel [5].

[2] This free-body argument uses statics only and is derived here for the eccentric cavity. The same balance, between the moment of the pressure on the distal cap and the moment of the elastomer stresses, is the basis of the quasi-static model of fibre-reinforced bending actuators in [3].

2.4 Euler–Bernoulli response

Under (A3), $\sigma_x = E\varepsilon_x$ with $\varepsilon_x = \varepsilon_0 + \kappa(y - \bar{y})$, and the Euler–Bernoulli moment–curvature relation [5, 6] gives

$$\kappa = \frac{M}{E I_{zz}} = \frac{\pi P R^2 r^2 r_p}{E\, I_{zz}\,(R^2 - r^2)}. \tag{2.9}$$

Here E is the small-strain modulus, $E = 3\mu = 6C_{10}$ for an incompressible solid.[3] Since M is uniform, the centreline is a circular arc and the tip angle is

$$\theta = \kappa\, l = \frac{\pi P R^2 r^2 r_p\, l}{E\, I_{zz}\,(R^2 - r^2)}, \tag{2.10}$$

linear in P. The transverse tip deflection of an arc of radius $\rho = 1/\kappa$ (constant-curvature geometry [10]) is

$$\delta = \frac{1 - \cos\theta}{\kappa} \;\approx\; \frac{\theta\, l}{2} \quad (\theta \ll 1). \tag{2.11}$$

In non-dimensional form, with $\hat{I} = I_{zz}/R^4$,

$$\theta = \frac{P}{E}\,\frac{l}{R}\, f\!\left(\frac{r}{R}, \frac{r_p}{R}\right), \qquad f = \frac{\pi (r/R)^2 (r_p/R)}{\hat{I}\,\big(1 - (r/R)^2\big)}. \tag{2.12}$$

Table 2.1. Beam model for the placeholder design $R = 10$, $r = 4$, $r_p = 3$, $l = 100\,\text{mm}$, $E = 0.6\,\text{MPa}$.

Section, eqs. (2.1)–(2.6)		Response at $P = 20\,\text{kPa}$	
A	$263.9\,\text{mm}^2$	F	$1.005\,\text{N}$
$\bar{y}$	$-0.571\,\text{mm}$	M	$3.590\,\text{N mm}$
I_{zz}	$7114.4\,\text{mm}^4$	κ	$8.41 \times 10^{-4}\,\text{mm}^{-1}$
e	$3.571\,\text{mm}$	θ	$4.82°$
$t_{\min},\ t_{\max}$	$3\,\text{mm}, 9\,\text{mm}$	δ (arc / small angle)	$4.203\,\text{mm} / 4.206\,\text{mm}$
$\hat{I},\ f$	$0.7114, 0.2523$		

2.5 The bare actuator: effect of the in-plane wall stresses

The pressure also loads the wall *in the plane of the section*: radial compression at the cavity surface and hoop tension around it. Assumption (A3) ignores these stresses. In linear isotropic elasticity they enter the axial strain through Poisson's ratio (generalised Hooke's law [6]),

$$\sigma_x = E\big[\varepsilon_0 + \kappa(y - \bar{y})\big] + \nu\,(\sigma_y + \sigma_z). \tag{2.13}$$

[3] For an isotropic linear-elastic solid $E = 2\mu(1 + \nu)$, so $E = 3\mu$ when $\nu = 1/2$ [6]. For the neo-Hookean and Yeoh models the small-strain shear modulus is $\mu = 2C_{10}$ [7, 8, 9].

The in-plane stresses need not be solved: away from the ends they satisfy $\partial_\beta \sigma_{\alpha\beta} = 0$ with a traction-free outer boundary and the traction $P\boldsymbol{m}$ on the cavity boundary. Using the identities $\partial_\beta(\sigma_{\alpha\beta} x'_\alpha) = \sigma_{\alpha\alpha}$ and $\partial_\beta(\sigma_{\alpha\beta} x'_\alpha x'_\gamma) - \partial_\beta(\sigma_{\gamma\beta} \frac{1}{2}|\boldsymbol{x}'|^2) = \sigma_{\alpha\alpha} x'_\gamma$ (coordinates $\boldsymbol{x}'$ centred on the cavity) and the divergence theorem,[4]

$$\int_A (\sigma_y + \sigma_z)\, dA = 2P\pi r^2, \qquad \int_A (\sigma_y + \sigma_z)(y - \bar{y})\, dA = 2P\pi r^2\, e, \tag{2.14}$$

for an outer boundary of any shape. Substituting eq. (2.13) into eq. (2.8) gives

$$\varepsilon_0 = (1 - 2\nu)\,\frac{P\pi r^2}{EA}, \qquad \kappa = (1 - 2\nu)\,\frac{M}{EI_{zz}}. \tag{2.15}$$

For silicone, $\nu \to 0.5$: **in the small-strain limit a bare eccentric-cavity actuator neither lengthens nor bends.** The thrust moment is present, but the Poisson effect of the in-plane stresses cancels it. For a concentric cavity the first result in eq. (2.15) reduces to the classical Lamé result for a closed-end thick-walled cylinder, whose axial strain is $(1 - 2\nu)Pr^2/[E(R^2 - r^2)]$ [6]: a closed incompressible tube does not lengthen under internal pressure. The second result is its bending counterpart. The beam result, eq. (2.10), therefore describes a hoop-reinforced actuator, not a bare one. This agrees with the common practice of restraining the radial expansion of pneumatic bending actuators with circumferential fibres [3, 4].

2.6 Asymmetric wall expansion and ballooning

Wall thickness. Along a ray from the cavity centre at angle ϕ from $+y$, the law of cosines in the triangle formed by the two centres and the point on the outer circle gives[5]

$$t(\phi) = \sqrt{R^2 - r_p^2 \sin^2\phi} - r_p \cos\phi - r, \tag{2.16}$$

with $t(0) = t_{\min}$ and $t(\pi) = t_{\max}$.

Second-order lengthening. For a thin closed neo-Hookean tube with hoop, axial and thickness stretches $\lambda_1 = 1 + a$, $\lambda_2 = 1 + b$, $\lambda_3 = 1/(\lambda_1\lambda_2)$ (incompressibility), the membrane stresses of an incompressible neo-Hookean sheet are $\sigma_\theta = \mu(\lambda_1^2 - \lambda_3^2)$ and $\sigma_x = \mu(\lambda_2^2 - \lambda_3^2)$ [7, 8], and a closed end gives $\sigma_x = \sigma_\theta/2$ (as for a thin-walled cylindrical pressure vessel

[4] Derived in this work from the in-plane equilibrium equations alone. Since $\partial_\beta \sigma_{\alpha\beta} = 0$, $\partial_\beta(\sigma_{\alpha\beta} x'_\alpha) = \sigma_{\alpha\alpha}$; integrating over the section and applying the divergence theorem gives $\int_A \sigma_{\alpha\alpha}\, dA = \oint x'_\alpha t_\alpha\, ds$, where $\boldsymbol{t}$ is the traction on the boundary. The outer boundary is traction-free. On the cavity boundary, $|\boldsymbol{x}'| = r$ and the pressure gives $\boldsymbol{t} = P\boldsymbol{x}'/r$, so $x'_\alpha t_\alpha = Pr$ and the integral is $2\pi r \cdot Pr = 2P\pi r^2$. The second identity, integrated in the same way, gives $\int_A \sigma_{\alpha\alpha} x'_\gamma\, dA = \oint \frac{1}{2} Pr\, x'_\gamma\, ds = 0$ on the cavity circle, i.e. $\int_A (\sigma_y + \sigma_z)(y - r_p)\, dA = 0$; adding $(r_p - \bar{y}) \int_A (\sigma_y + \sigma_z)\, dA$ gives the second result with $e = r_p - \bar{y}$.

[5] The point on the outer circle at distance s from the cavity centre along the ray satisfies $R^2 = r_p^2 + s^2 + 2r_p s\cos\phi$ (law of cosines); solving for s and subtracting r gives the wall thickness.

[5]), i.e. $2\lambda_2^2 = \lambda_1^2 + \lambda_3^2$. Expanding:[6] at first order $b = 0$ (no axial strain, in agreement with eq. 2.15); at second order $6b = 4a^2$, so

$$\varepsilon_{\text{axial}} \approx \tfrac{2}{3}\,\varepsilon_{\text{hoop}}^2. \tag{2.17}$$

The thin wall has the larger hoop strain, so it lengthens more and the actuator bends toward the thick wall. This bending mechanism is real but quadratic in P. A linear estimate of it (a curvature proportional to P/t) is therefore not valid: it rests on the same linear Poisson coupling that, by eq. (2.15), gives no net bending at all.

Ballooning. A rubber tube under internal pressure can reach a pressure maximum (a limit point): beyond it the wall thins faster than the hoop stress can rise and the tube inflates at falling pressure, often as a local bulge [11]. Under pressure control there is no equilibrium above the maximum. In the eccentric actuator the thin wall has the largest hoop strain and is expected to reach this state first. The limit pressure of the eccentric section was not computed in this work (the bare single-cavity COMSOL run of Chapter 4 ends at 90 kPa, the last pressure requested). A real silicone stiffens at large stretch (described, for example, by the Yeoh coefficients $C_{20}, C_{30} > 0$ [9]), and strain stiffening changes the inflation response beyond the pressure maximum [11]; the measured material is needed before any limit pressure is used for design.

2.7 Model hierarchy

Table 2.2. Model tiers. Tiers 1–3 share assumption (A3) and describe the hoop-reinforced actuator; the bare actuator needs Tier 4.

Tier	Content	Applies to
1	Linear Euler–Bernoulli, eqs. (2.9)–(2.11)	hoop-reinforced, small P/E
2	Tier 1 curvature with arc kinematics and axial stretch: $\rho = l(1+\varepsilon_0)/\theta$, $x_{\text{tip}} = \rho\sin\theta$, $\delta = \rho(1-\cos\theta)$; constant-curvature kinematics [10]	hoop-reinforced; form used for control
3	Hyperelastic section equilibrium: plane sections, uniaxial hyperelastic fibres, eq. (2.8) solved numerically (cf. [3])	hoop-reinforced, larger P
4	Nonlinear three-dimensional FE in COMSOL (Chapter 3)	bare and reinforced
5	Experiment	material constants, validation

[6] Derived in this work. With $\lambda_3^2 = (1+a)^{-2}(1+b)^{-2} = 1 - 2a - 2b + 3a^2 + 3b^2 + 4ab + \ldots$, the condition $2\lambda_2^2 = \lambda_1^2 + \lambda_3^2$ becomes $2 + 4b + 2b^2 = 2 - 2b + 4a^2 + 3b^2 + 4ab + \ldots$. The first-order terms give $6b = 0$; with b of order a^2, the second-order terms give $6b = 4a^2$.

CHAPTER 3

METHODOLOGY: FINITE-ELEMENT MODELS IN COMSOL

The analytical model of Chapter 2 rests on assumptions that the FE models do not make: they use finite strains, a nonlinear (hyperelastic) material, the full three-dimensional stress state in the wall, and a pressure that follows the deforming cavity surface. This chapter describes how the models were built and solved.

3.1 Software and workflow

All FE models were built in COMSOL Multiphysics 6.4 with the Structural Mechanics Module [12, 13]. Instead of building the models by hand in the graphical interface, each model was written as a Java program that uses the COMSOL application programming interface (API). One program creates, in order, the parameters, the geometry, the boundary selections, the physics (material, loads and supports), the mesh and the study, runs the solver, evaluates the output quantities at every pressure step and prints them as text lines. The programs were compiled with `comsolcompile` and run without the graphical interface with `comsolbatch`; a short Python script reads the printed lines into CSV files, from which the tables and graphs of Chapter 4 are made. Each run also saves the solved model (`.mph` file), which can be opened in the COMSOL interface to inspect the solution. Working this way makes every model parametric (a different design is a different set of numbers) and every result reproducible from its script. The files are listed in Appendix A.

Two models were built:

1. the *single-cavity actuator* of Chapter 2, bare and with a hoop sleeve, to test the analytical model; and
2. the *tapered four-chamber actuator*, bare and sleeved, with variants of the chamber shape.

3.2 Geometry

Single-cavity actuator. The placeholder design of Chapter 2: a cylinder of radius $R = 10\,\text{mm}$ and length $l + t_{\text{cap}} = 105\,\text{mm}$ along x, from which a cylinder of radius $r = 4\,\text{mm}$, centred at $y = r_p = 3\,\text{mm}$, is subtracted from $x = 0$ to $x = l = 100\,\text{mm}$. The cavity is therefore open at the base (the air inlet) and closed by a 5 mm solid cap at the tip. Because the geometry and

the load are symmetric about the bending plane $z = 0$, only the half $z \geq 0$ is modelled; it is cut from the full body by intersecting it with a block.

Tapered four-chamber actuator. Length $L = 140\,\text{mm}$, outer radius tapering linearly from $R_0 = 9.5\,\text{mm}$ at the base to $R_t = 3.8\,\text{mm}$ at the tip (a cone). Four chambers of angular width 56°, separated by 34° ribs, run from $x = 4$ to 136 mm. At $x = 4\,\text{mm}$ each chamber spans the radii $a = 5.0\,\text{mm}$ to $b = 7.5\,\text{mm}$ (the outer radius there is 9.34 mm, so the roof is 1.84 mm thick), and all chamber radii decrease linearly to 0.4 times these values at $x = 136\,\text{mm}$ (Figures 3.1 and 3.2). The 4 mm plug at the base and the 4 mm cap at the tip are solid. Each chamber is built as the region between two coaxial cones (the inner and outer chamber surfaces), bounded by two planes through the axis, and is subtracted from the outer cone. For the variant with rounded chamber corners, the cross-section of a chamber (an annular sector) was drawn in the plane $x = 4\,\text{mm}$, its four corners were filleted with radius 0.6 mm, and the section was extruded to $x = 136\,\text{mm}$ with a scale factor of 0.4. Because all chamber radii scale by the same factor, this extrusion reproduces the same linear taper, and the fillet radius decreases to 0.24 mm at the tip. The four-chamber actuator has no symmetry plane that is convenient for all bending directions, so the full body is modelled.

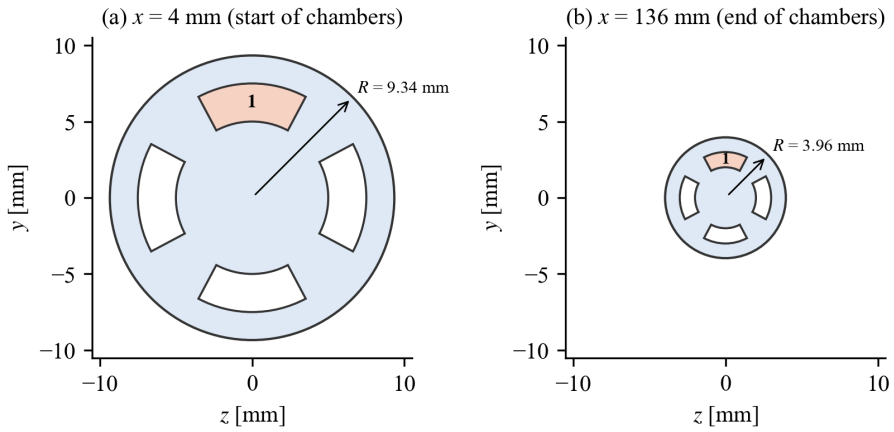
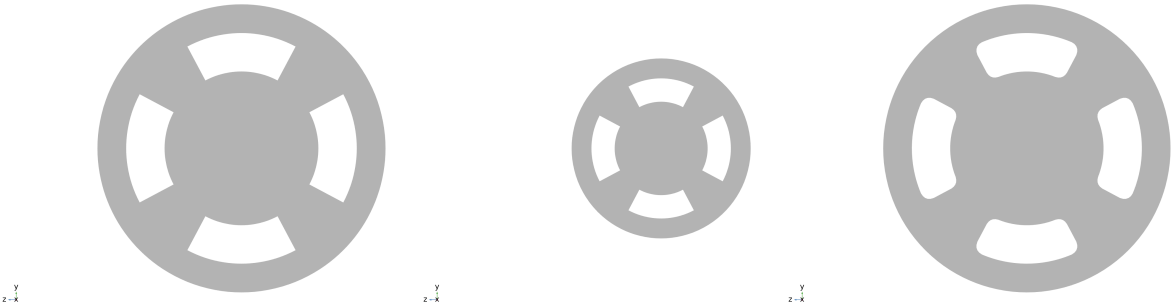


Figure 3.1. Cross-section of the tapered four-chamber actuator at the two ends of the chambers (true scale). Chamber 1 (shaded) is pressurised; the others are vented.



(a) $x = 20\,\mathrm{mm}$ **(b)** $x = 100\,\mathrm{mm}$ **(c)** $x = 20\,\mathrm{mm}$, filleted

Figure 3.2. Sections of the COMSOL geometry of the four-chamber actuator (undeformed), viewed along the axis from the tip: (a), (b) original chambers at two positions, drawn at the same scale, showing the taper; (c) the variant with 0.6 mm fillets at the chamber corners. Chamber 1, the pressurised one, is at the top ($+y$).

Selection of boundaries. Loads and supports act on boundary faces, which must be identified reliably. In the single-cavity model they are selected with boxes and cylinders in space (for example, all faces lying within 0.1 mm of the outer radius). In the four-chamber model the faces are classified from their geometry alone: for every face the script computes its area, the mean axial position, the mean direction from the axis and the root-mean-square distance from the outer cone. Faces at $x = 0$ are the base, faces at $x = L$ are the tip, faces within 0.5 mm (r.m.s.) of the outer cone are the outer skin (the r.m.s. distance of a true skin face is about 0.06 mm, caused by the faceting of the curved surface, while chamber faces are more than 1.5 mm away), and every remaining face is assigned to the chamber in whose direction it lies. The classification is printed and was checked for every run.

3.3 Material model

The silicone is modelled as a nearly incompressible neo-Hookean solid [7, 8]. With the deformation gradient $\boldsymbol{F}$, $J = \det \boldsymbol{F}$ and the isochoric invariant $\bar{I}_1 = J^{-2/3}\,\mathrm{tr}(\boldsymbol{F}^{\mathsf{T}}\boldsymbol{F})$, the strain-energy density is

$$W = \frac{\mu}{2}\big(\bar{I}_1 - 3\big) + \frac{\kappa_b}{2}\big(J - 1\big)^2, \tag{3.1}$$

which is COMSOL's neo-Hookean model with the "nearly incompressible" option and a quadratic volumetric energy [13]. The bulk modulus is set from Poisson's ratio, $\kappa_b = 2\mu(1+\nu)/[3(1-2\nu)]$ with $\nu = 0.4995$, so that $\kappa_b \approx 1000\,\mu$ and the volume of the material hardly changes. A nearly incompressible material cannot be computed accurately with displacements alone, because the elements become far too stiff in volume (volumetric locking). COMSOL's mixed ("p-") formulation was therefore used, in which an auxiliary pressure field is solved for together with the displacements [8, 13].

The moduli are placeholders. The single-cavity model uses $E = 0.6\,\mathrm{MPa}$ ($\mu = E/3 = 0.2\,\mathrm{MPa}$), the value used in Chapter 2. The four-chamber model uses $E = 0.07\,\mathrm{MPa}$

($\mu = E/[2(1+\nu)] = 23.3\,\mathrm{kPa}$), an assumed value of the order of the 100 % modulus of a soft platinum-cure silicone (69 kPa for Ecoflex 00-30 [14]); it has not been measured. For a neo-Hookean material a change of modulus only rescales the pressure (the response depends on P/μ), so these choices do not affect the shape of the results.

3.4 Supports and loads

- **Base.** All displacements of the base face ($x = 0$) are zero, which represents an actuator clamped at its base.
- **Symmetry** (single-cavity model only). The displacement normal to the plane $z = 0$ is zero.
- **Cavity pressure.** A gauge pressure P acts on every face of the pressurised cavity: the cavity wall and the inner face of the tip cap, and, for the four-chamber model, every face of chamber 1. It is applied as a *follower* pressure, which acts normal to the deformed surface and on its deformed area [13], as the pressure of a gas or liquid does. The other three chambers of the four-chamber actuator are vented (no load). Gravity is neglected.

Hoop sleeve. A sleeve of circumferential fibres around the outer surface is modelled as a surface energy on the outer skin, added through COMSOL's *weak contribution* feature, which adds a user-written term to the weak (virtual-work) form of the equations [12]. In reference coordinates (X, Y, Z) the hoop direction is the unit vector $\boldsymbol{t} = (0, -Z, Y)/\sqrt{Y^2 + Z^2}$, and the Green strain of the surface along $\boldsymbol{t}$ is [8]

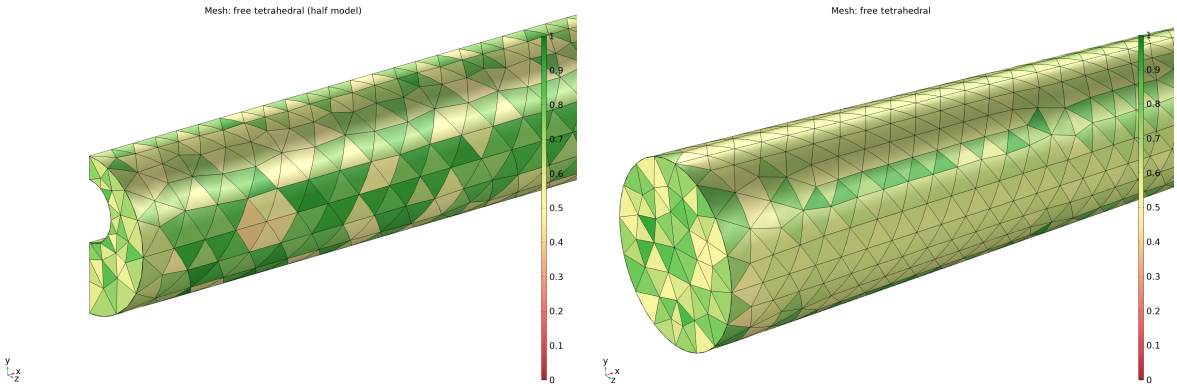
$$e_h = \tfrac{1}{2}\left(\lambda_h^2 - 1\right) = \boldsymbol{t} \cdot \left(\nabla \boldsymbol{u}\right)\boldsymbol{t} + \tfrac{1}{2}\left|\left(\nabla \boldsymbol{u}\right)\boldsymbol{t}\right|^2, \qquad \lambda_h = |\boldsymbol{F}\boldsymbol{t}|, \tag{3.2}$$

where $\boldsymbol{u}$ is the displacement. Only derivatives tangential to the surface are needed, because $\boldsymbol{t}$ lies in the surface. The sleeve stores the energy $\frac{1}{2}k_s e_h^2$ per unit reference area, i.e. the weak contribution $-k_s\, e_h\, \delta e_h$ integrated over the outer skin. The sleeve therefore resists hoop stretching only; it has no axial and no bending stiffness and is perfectly bonded to the silicone. The stiffness $k_s = 100\,ER$ (R at the base for the tapered actuator) is a penalty value, chosen to be much larger than the hoop membrane stiffness of the silicone wall (of order Et with wall thickness $t < R$), so that the sleeve is nearly inextensible. Because $\boldsymbol{t}$ is tangent to a cone as well as to a cylinder, the same formulation applies to the tapered actuator.

3.5 Mesh

Both models use a free tetrahedral mesh generated automatically with COMSOL's "Fine" size preset, with quadratic displacement shape functions. The four-chamber mesh has 18 044

elements (20 946 for the deeper chambers and 36 362 with fillets, which need small elements to resolve the rounded corners). Figure 3.3 shows the surface of the two meshes near the base.



(a) single-cavity actuator (half model) **(b)** four-chamber actuator

Figure 3.3. COMSOL meshes near the base ($x = 0$ at the left end). (a) is the half model, cut at the symmetry plane $z = 0$; the notch at the left end is the cavity opening at the base. The elements are coloured by COMSOL's element-quality measure, from 0 (degenerate) to 1 (ideal).

3.6 Solution procedure

A stationary study with geometric nonlinearity (large deformation) was used. The pressure is increased in a parametric sweep, each step starting from the converged solution of the previous one: 0 kPa to 110 kPa in steps of 10 kPa for the sleeved single-cavity actuator, 0 kPa to 90 kPa for the bare one, and 0 kPa to 100 kPa for the four-chamber actuator with steps of 2 kPa to 2.5 kPa below 30 kPa. Each step is solved with COMSOL's Newton-type nonlinear solver; when a step does not converge, the solver tries smaller intermediate pressure increments. When even these fail, the run stops and all steps converged so far are kept; the last of them is reported as the "last converged step". On the project computer one four-chamber run took between 3 min (bare) and 57 min (with fillets).

3.7 Quantities extracted

Single-cavity actuator. The tip angle is the rotation of the line joining the two outer-surface points at the end of the cavity, $(x, y, z) = (100, \pm 10, 0)$ mm: $\theta = \arctan(\Delta x / \Delta y)$ from their deformed positions. The tip deflection δ is minus the mean y-displacement of the two points.

Four-chamber actuator. Tip rotation: angle of the deformed material line joining the points $y = \pm 1.5$ mm at $x = 139.8$ mm, followed continuously from step to step so that rotations beyond 90° are counted correctly. Deformed centreline: 29 points on the axis, which lies in the solid core. Section change: deformed distance between the two outer-skin points on the y axis at $x = 70$ mm, divided by the original diameter. The peak von Mises stress and the

smallest local volume ratio J over the whole body are recorded at every step as indicators of local distortion.

3.8 Mesh check

With the “Fine” mesh preset replaced by “Finer”, the tip rotation of the sleeved four-chamber actuator at 10 kPa changes by 1.2 % (47.1° to 47.7°) and the tip deflection by 0.9 %. The check was not repeated at higher pressure, and no mesh check was made for the single-cavity model.

3.9 Beam estimate for the tapered actuator

For comparison, eq. (2.9) was integrated over the tapered section. A chamber is an annular sector of half-angle α between radii $a(x)$ and $b(x)$, with area $A_c = \alpha(b^2 - a^2)$ and centroid at $e = \frac{2\sin\alpha}{3\alpha}\frac{b^3-a^3}{b^2-a^2}$ from the axis.[1] The free-body argument of Section 2.3 gives $M(x) = PA_c(x)e(x)$ at each section, and the section with four equal voids has its centroid on the axis and $I = \pi R^4/4 - \alpha(b^4 - a^4)$ about it,[2] so

$$\theta_{\text{beam}} = \frac{P}{E}\int_{x_s}^{x_e} \frac{A_c(x)\,e(x)}{I(x)}\,dx. \tag{3.3}$$

Like eq. (2.10), this applies to a hoop-reinforced actuator at small P/E; the solid plug and cap are treated as rigid.

[1] By integration in polar coordinates: $A_c = \int_{-\alpha}^{\alpha}\int_a^b \rho\,d\rho\,d\varphi$ and $A_c e = \int_{-\alpha}^{\alpha}\int_a^b \rho\cos\varphi\,\rho\,d\rho\,d\varphi = \frac{2}{3}\sin\alpha\,(b^3 - a^3)$; see the tables of centroids in [5].

[2] A sector contributes $\iint \rho^2\sin^2\varphi\,\rho\,d\rho\,d\varphi$; for four sectors 90° apart the angular integrals add up to 4α, so the four voids remove $\frac{1}{4}(b^4 - a^4)\cdot 4\alpha$ from the solid disc.

CHAPTER 4

RESULTS

4.1 Single-cavity actuator

Table 4.1 and Figure 4.1 compare the COMSOL results with the beam model, eqs. (2.10) and (2.11), for the placeholder design ($E = 0.6\,\text{MPa}$, $\mu = 0.2\,\text{MPa}$).

Table 4.1. Single-cavity actuator, $E = 0.6\,\text{MPa}$: tip angle θ and tip deflection δ from COMSOL, and the ratio of the COMSOL tip angle to the beam model. The bare run ends at 90 kPa, the last pressure requested. For another neo-Hookean silicone the pressure scales with μ.

		Bare			Hoop-reinforced (sleeved)		
P [kPa]	θ_{beam} [°]	θ [°]	FE/beam	δ [mm]	θ [°]	FE/beam	δ [mm]
10	2.41	0.263	0.11	0.22	2.43	1.01	2.12
20	4.82	0.761	0.16	0.64	5.06	1.05	4.44
40	9.64	2.78	0.29	2.37	11.0	1.15	9.83
60	14.5	7.12	0.49	6.12	18.2	1.26	16.4
80	19.3	16.6	0.86	14.4	26.7	1.39	24.3
90	21.7	26.0	1.20	22.6	31.6	1.46	28.9
100	24.1	–	–	–	37.0	1.54	34.0
110	26.5	–	–	–	43.0	1.62	39.5

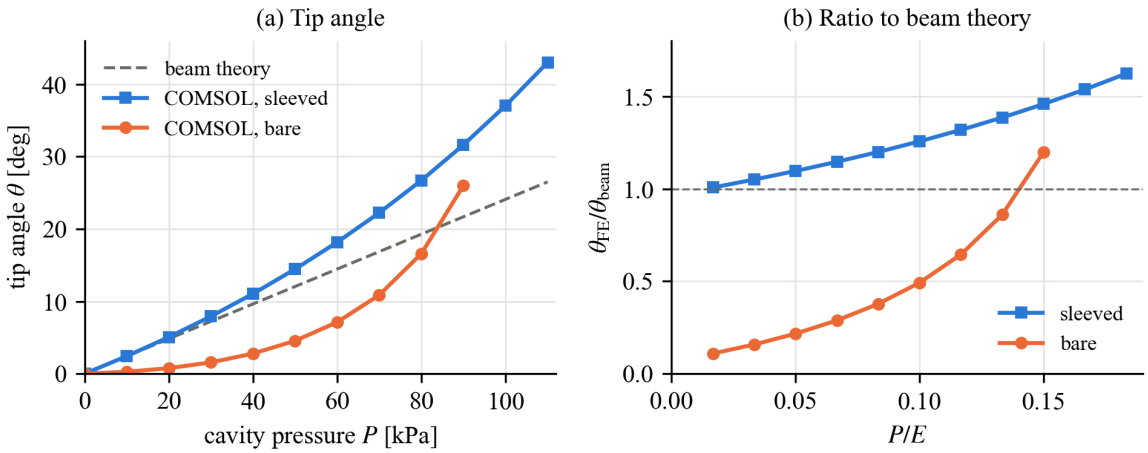


Figure 4.1. Single-cavity actuator: (a) tip angle against pressure from the beam model, eq. (2.10), and from COMSOL, bare and sleeved; (b) ratio of the COMSOL tip angle to the beam model against P/E.

Hoop-reinforced actuator. The response is smooth up to 110 kPa, the last pressure requested, and every requested step converged. At small pressure it agrees with the beam model: the ratio of the tip angles is 1.01 at $P/E = 0.017$ (10 kPa) and 1.05 at $P/E = 0.033$ (20 kPa). Since eq. (2.10) and the FE model share only the geometry, the modulus and the pressure, this agreement is also an independent check of the FE model set-up (geometry, material, follower load and the way the tip angle is measured). At higher pressure the FE model bends more than the beam model, by 26 % at $P/E = 0.1$ and 62 % at 110 kPa. The beam model uses the undeformed section and a linear material, whereas the FE model follows the change of geometry with pressure; the separate contributions of these effects were not determined.

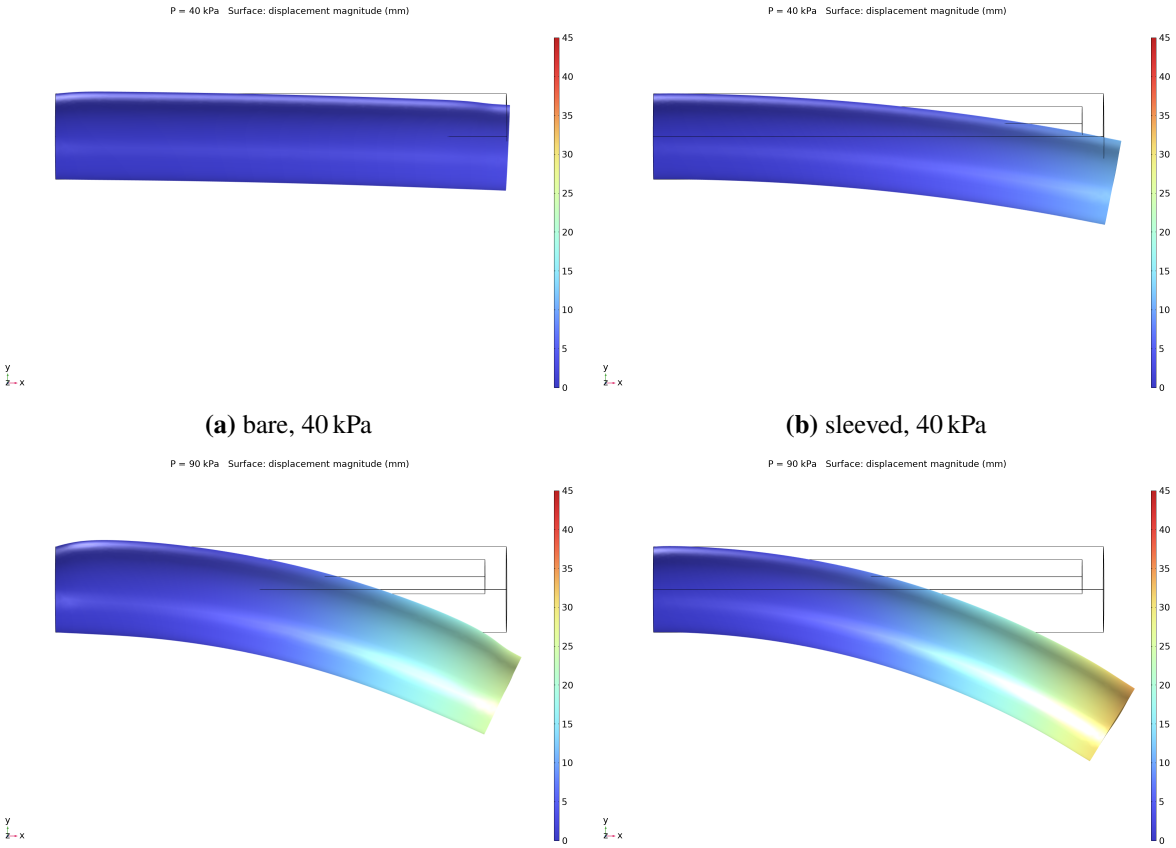
Bare actuator. The bare actuator bends far less than the beam model at small pressure: 11 % of it at 10 kPa and 16 % at 20 kPa. This is the reduction predicted by eq. (2.15). The remaining bending is not zero; these runs do not separate the second-order mechanism of eq. (2.17) from the effect of the two ends, where the clamp and the cap restrain the wall and the in-plane stress state of Section 2.5 does not hold. The tip angle grows faster than linearly with pressure (by a factor of 2.9 from 10 to 20 kPa and 3.7 from 20 to 40 kPa), as expected of a mechanism that is quadratic in the hoop strain. The FE and beam curves cross between 80 and 90 kPa; this is a coincidence of two different mechanisms, not an agreement. Between 80 and 90 kPa the deflection increases by 8.2 mm, against 5.0 mm between 70 and 80 kPa, so the response is steepening, as expected on the approach to ballooning (Section 2.6). The run ends at 90 kPa, so the limit pressure itself was not reached.

Deformed shapes. Figure 4.2 shows the two actuators at 40 kPa and 90 kPa with the same colour scale, and Figure 4.3 the section at mid-length at 90 kPa. At 40 kPa the bare actuator is still almost straight (2.8°), while the sleeved one has bent by 11°. At 90 kPa the cavity of the bare actuator has grown far more than that of the sleeved one, and the von Mises stress is highest in the material around the cavity; in the sleeved actuator the outer boundary of the section stays circular, because the sleeve prevents it from stretching in the hoop direction.

4.2 Four-chamber actuator

4.2.1 Bare and sleeved actuator

Table 4.2 and Figures 4.4–4.7 give the results of the four-chamber model of Chapter 3 ($E = 0.07$ MPa, $\mu = E/(2(1+\nu)) = 23.3$ kPa).



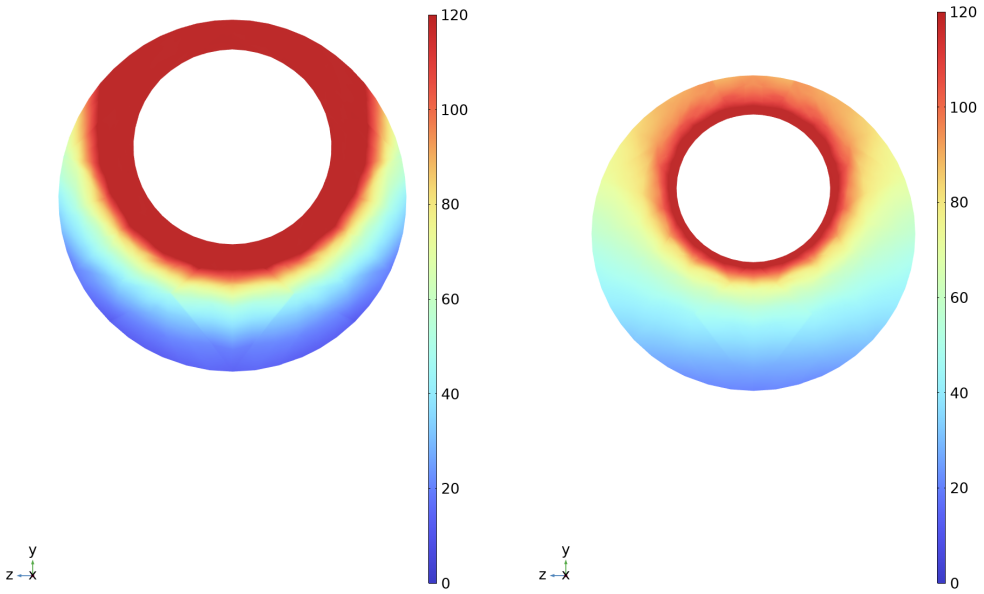
(a) bare, 40 kPa

(b) sleeved, 40 kPa

(c) bare, 90 kPa

(d) sleeved, 90 kPa

Figure 4.2. Single-cavity actuator, $E = 0.6$ MPa: deformed shape viewed normal to the bending plane (COMSOL, true scale; the half model is mirrored about $z = 0$ for display). Colour: displacement magnitude [mm], same scale (0–45 mm) in all four panels. The thin lines show the undeformed body; the base is clamped at the left.



(a) bare **(b)** sleeved

Figure 4.3. Single-cavity actuator at $P = 90$ kPa: deformed section at $x = 50$ mm, viewed along the axis from the tip (COMSOL, true scale). Colour: von Mises stress [kPa], same scale in both panels; values above 120 kPa are shown in the top colour. Undeformed, the section is a circle of radius 10 mm with a cavity of radius 4 mm centred 3 mm above the axis.

Table 4.2. Four-chamber actuator, chamber 1 pressurised, $E = 0.07$ MPa: tip rotation θ [°]. "Sleeved" has the original chambers; "filleted" adds 0.6 mm fillets to the chamber corners (0.24 mm at the tip); "deeper" has chambers 5.0 mm to 1.5 mm deep and 70° wide. Beam: eq. (3.3). – : no converged solution.

P [kPa]	bare	sleeved	sleeved, filleted	sleeved, deeper	beam (sleeved)
2	0.73	6.06	4.83	8.33	5.74
4	3.06	13.7	11.2	19.5	11.5
6	7.89	22.8	19.1	34.0	17.2
8	17.2	33.8	28.7	55.3	22.9
10	36.6	47.1	40.6	–	28.7
12.5	–	68.0	59.4	–	35.8
15	–	95.3	84.6	–	43.0
last converged step	11.7 kPa	20 kPa	21 kPa	8.5 kPa	

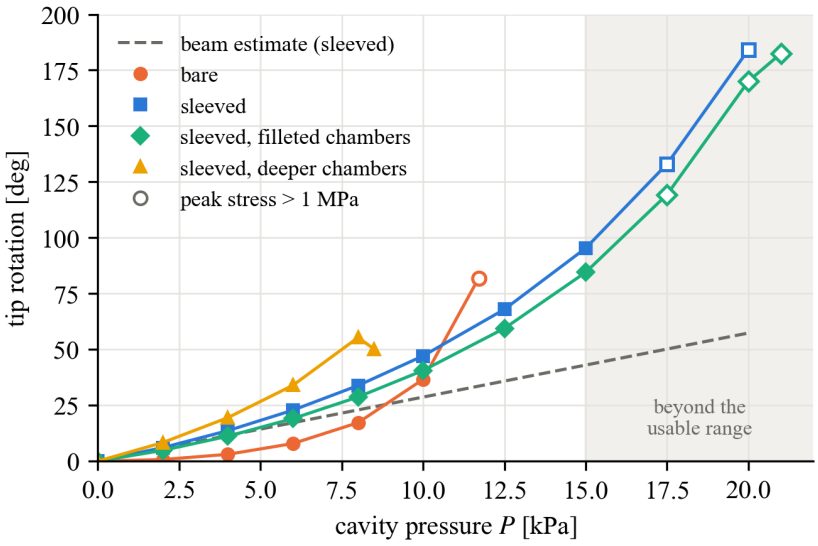


Figure 4.4. Four-chamber actuator with real pressure: tip rotation against pressure. Open symbols: steps where the computed peak von Mises stress exceeds 1 MPa. The shaded band lies beyond the range in which results are used (Section 4.2.2).

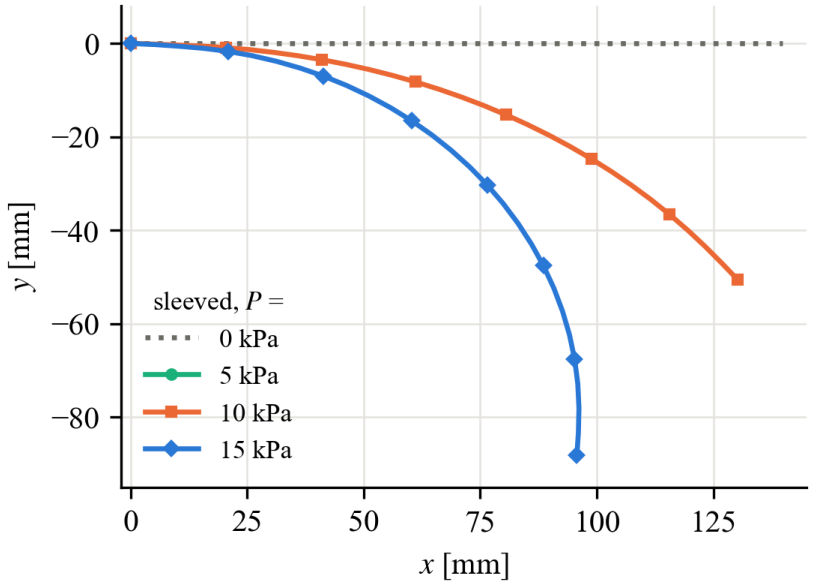
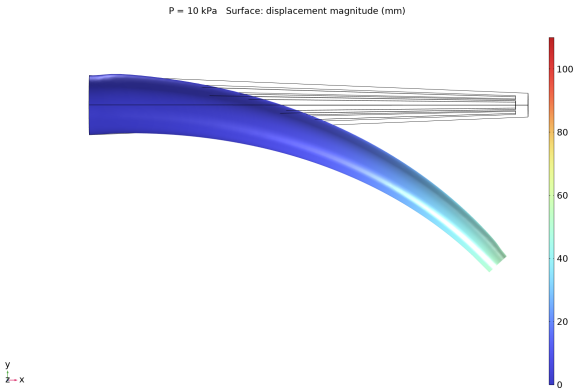
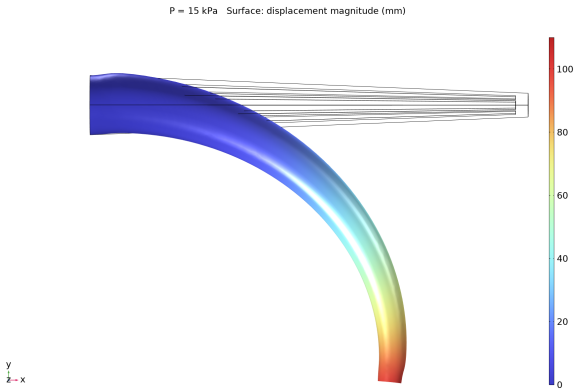


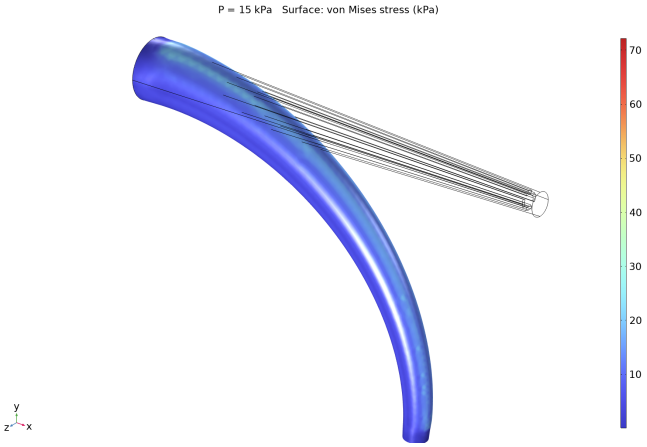
Figure 4.5. Deformed centreline of the sleeved four-chamber actuator in the bending plane (true scale).



(a) 10 kPa, displacement [mm]

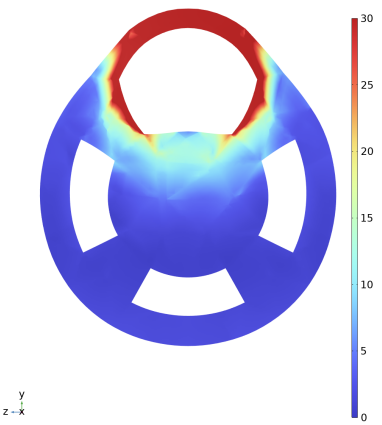


(b) 15 kPa, displacement [mm]

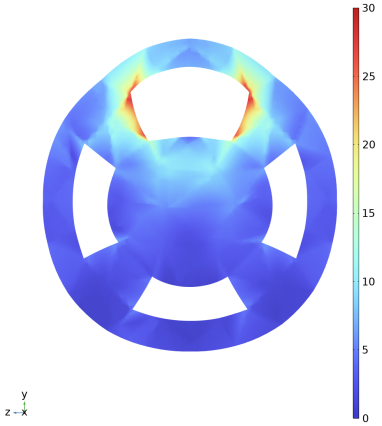


(c) 15 kPa, von Mises stress on the outer surface [kPa], oblique view

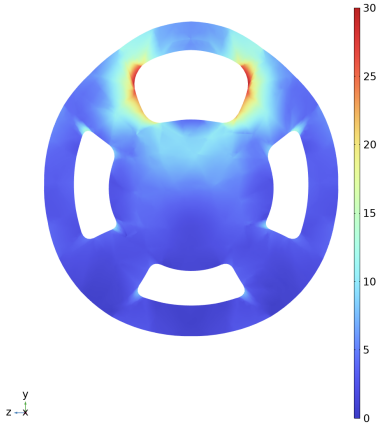
Figure 4.6. Sleeved four-chamber actuator, $E = 0.07$ MPa (COMSOL, true scale; the thin lines show the undeformed body). (a), (b) viewed normal to the bending plane, base clamped at the left. The colour scales differ between panels; quantitative values are in Table 4.2 and Figure 4.5. In (c) only the outer surface is shown; the highest stresses lie inside the walls, at the chamber corners (Section 4.2.2).



(a) bare



(b) sleeved



(c) sleeved, filleted

Figure 4.7. Four-chamber actuator at $P = 10$ kPa: deformed section at $x = 40$ mm, viewed along the axis from the tip (COMSOL, true scale; chamber 1 at the top is pressurised). Colour: von Mises stress [kPa], same scale in all three panels; values above 30 kPa are shown in the top colour. Compare the undeformed sections in Figure 3.2.

Bare. The bare actuator bends little at first (0.73° at 2 kPa) and then increasingly fast, as the bare single-cavity actuator does. The section at mid-length grows by 13.8 % at 10 kPa, and the solver stops at 11.7 kPa, i.e. $P/\mu = 0.50$, which is consistent with the onset of ballooning of the chamber roof (Section 2.6); the limit pressure was not located precisely. The last step is not used: its computed stress is unphysical. Figure 4.7(a) shows the section at 10 kPa: the roof of chamber 1 has bulged far beyond the original outer surface, and the stress is highest in the roof and in the two ribs next to it.

Sleeved. With the hoop sleeve the response starts linearly and close to the beam estimate (6.06° against 5.74° at 2 kPa) and then bends more than the estimate, as the single-cavity sleeved actuator does: 47.1° at 10 kPa and 95.3° at 15 kPa. The section at mid-length grows by only 4.6 % at 15 kPa. In the section of Figure 4.7(b) the roof of chamber 1 stays close to the original outer circle; the chamber widens, and the stress concentrates at its corners, where the roof meets the ribs.

Chamber geometry. Fillets on the chamber corners add material and reduce the bending by about 10 % (84.6° at 15 kPa); at 10 kPa the deformed section and its stress pattern are similar to those without fillets (Figure 4.7(c)). Deeper and wider chambers bend more at low pressure (55.3° at 8 kPa) but the rotation then decreases and the mid-length section shrinks: the thin ribs and the thin core (radius 2.5 mm to 1.5 mm) collapse, and the solution stops at 8.5 kPa. The beam estimate, which assumes the section keeps its shape, had predicted that deeper chambers would be better; the FE result shows that they are not.

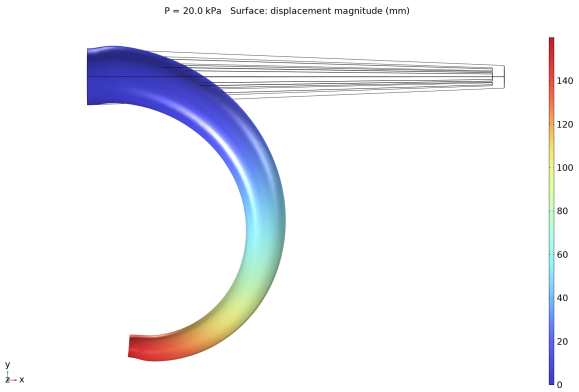
4.2.2 Range of the results

Above 15 kPa the sleeved models still converge (to 20 kPa, and 21 kPa with fillets) and the centreline continues to curl (to about 180°), but two quantities grow rapidly:

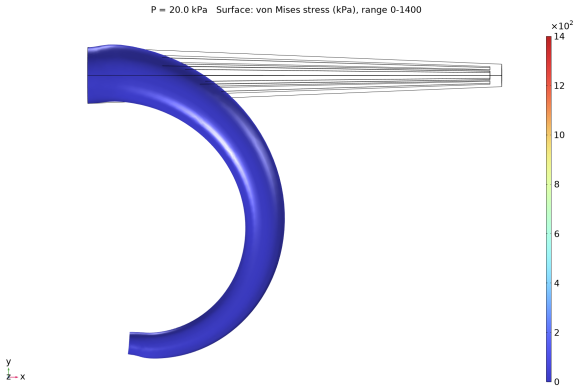
- the computed peak von Mises stress, located at the junction between the roof of the pressurised chamber and its side walls: in the sharp-cornered model 0.25 MPa at 15 kPa, 1.04 MPa at 17.5 kPa and 53 MPa at 20 kPa;
- the local element distortion: the smallest local volume ratio J falls from 0.92 at 2 kPa to 0.30 at 15 kPa and 0.01 at 20 kPa, while the volume average of J stays at 1.0000.

Fillets do not remove the stress growth, so it is not a sharp-corner singularity. Whether it is a physical instability of the chamber wall junction or a numerical breakdown of the elements cannot be decided from these runs. Results are therefore used only up to 15 kPa; of these, only the 10 kPa point has been mesh-checked. For comparison, the tensile strength of Ecoflex 00-30 is about 1.4 MPa [14].

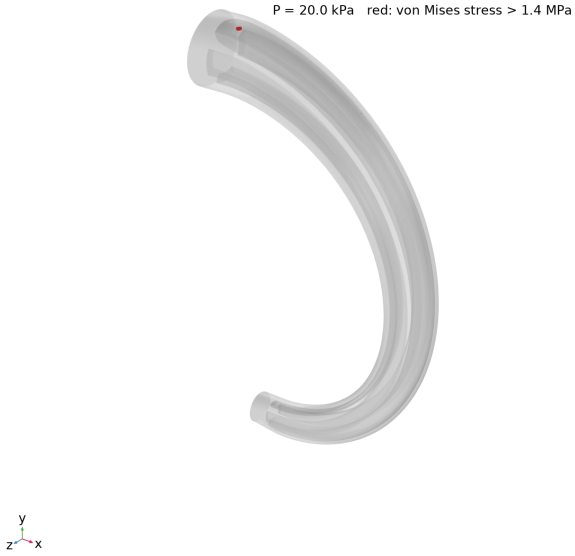
Bending to 180°: beyond the safe limit of the material. Figure 4.8 shows the sleeved actuator at its last converged step, $P = 20\,\text{kPa}$, where the tip has rotated by 184° and the actuator has curled into a hook. This shape lies *beyond the safe limit of the material*. At this step the computed peak von Mises stress is 53 MPa, about 38 times the tensile strength of Ecoflex 00-30 (about 1.4 MPa [14]), the silicone whose stiffness is assumed in this model; already at 17.5 kPa (133°) the peak, 1.04 MPa, is close to that strength. On the outer surface the stress stays well below 1.4 MPa (Figure 4.8(b)); the material above 1.4 MPa is a small volume inside the actuator, near the base end of chamber 1 (Figure 4.8(c), (d)). Whether this local stress is physical or a numerical artefact is not resolved (see above), but as computed it exceeds the strength of the silicone, so the material would be expected to fail there before the actuator reached 180°. These figures therefore show what the model does, not a usable operating point; all results quoted elsewhere in this report are for $P \leq 15\,\text{kPa}$.



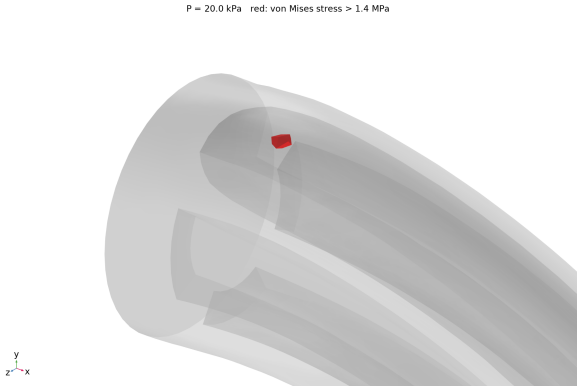
(a) displacement magnitude [mm]



(b) von Mises stress on the outer surface [kPa], scale 0 kPa to 1400 kPa



(c) oblique view; red: von Mises stress > 1.4 MPa



(d) close-up of the base end of (c)

Figure 4.8. Sleeved four-chamber actuator, $E = 0.07$ MPa, at $P = 20$ kPa, the last converged step (tip rotation 184°; COMSOL, true scale; the thin lines in (a), (b) show the undeformed body; base clamped at the left). In (c) and (d) the body is drawn transparent and the elements whose von Mises stress exceeds 1.4 MPa, the tensile strength of Ecoflex 00-30 [14], are shown in red. The computed peak is 53 MPa. **This state is beyond the safe limit of the material** and beyond the range in which the results are used (Section 4.2.2).

CHAPTER 5

DISCUSSION

5.1 How the actuator bends

The analysis separates two mechanisms that the simple beam picture combines. The eccentric thrust moment, eq. (2.7), is always present and uniform, but in a bare incompressible actuator its effect is cancelled at first order by the in-plane wall stresses, eq. (2.15). What remains is the second-order lengthening of the thin wall, eq. (2.17), which starts quadratically and ends at a ballooning limit. A bare actuator is therefore weak at low pressure and unstable near the limit. The COMSOL model of the single-cavity actuator shows this weak, steepening response; the limit pressure itself was not computed. With a hoop reinforcement the in-plane cancellation is removed, the beam formula becomes the correct first-order model, and the response is smooth up to the highest pressure simulated. The same picture holds for the tapered four-chamber actuator: bare, it balloons at $P/\mu \approx 0.5$; sleeved, it bends smoothly and starts close to the beam estimate.

5.2 Choice of material

For a neo-Hookean material the response depends on P/μ only. A softer silicone therefore gives the same curve at a proportionally lower pressure, and a stiffer silicone at a higher pressure; the material alone does not change how far a given geometry can bend. Any ballooning limit of a bare actuator, and the bending reached before the stress growth of Section 4.2.2, are fixed fractions of μ for a given geometry. A real silicone stiffens at large stretch, which a Yeoh fit to measured data would capture; this may raise the ballooning limit and delay the stress growth, and needs to be checked with the measured material.

5.3 Checks on the FE models

No second FE code and no experiment were available, so the FE results rest on the following checks:

- at small pressure the sleeved single-cavity model agrees with the closed-form beam result within 1 % (10 kPa), which tests the geometry, the material constants, the follower load and the measurement of the tip angle together;
- the bare single-cavity model shows the large reduction of bending predicted by eq. (2.15);

- the sleeved four-chamber model agrees with the tapered beam estimate, eq. (3.3), within 6 % at 2 kPa;
- the four-chamber tip rotation changes by 1.2 % with a finer mesh at 10 kPa;
- the boundary classification of the four-chamber model is printed and checked for every run, so that the pressure is known to act on the faces of chamber 1 only.

Every FE result in this report uses the real cavity pressure, applied uniformly to every face of the pressurised cavity, and nearly incompressible silicone, so that the pressure axis of every figure is a physical pressure that can be set in an experiment. What these checks do not cover is listed in Section 5.5.

5.4 Towards grasping

The long-term aim is a finger that curls around an object. With the present four-chamber geometry and a hoop sleeve, about 85°–95° of tip rotation is obtained within the range of trusted results, which is less than needed to wrap an object. Contact with an object has not been simulated with the physical model, so no grasping force is reported. To get there, the following are needed: resolving the stress growth of Section 4.2.2 (finer local mesh or hexahedral wall elements), possibly thicker chamber side walls or a longer actuator, measured material constants, and contact modelling with a stiff penalty or an augmented Lagrangian formulation, with the wrap measured as the arc actually in contact.

5.5 Limitations

- All dimensions and moduli are placeholders; $E = 0.07\,\mathrm{MPa}$ for the four-chamber actuator is assumed.
- Neo-Hookean material; strain stiffening, viscoelasticity and the Mullins effect are not modelled.
- The sleeve is ideal: inextensible in hoop, no axial stiffness, perfectly bonded.
- Only pitch bending (one chamber) of the four-chamber actuator has been computed with real pressure.
- The four-chamber mesh check covers 10 kPa only; the single-cavity models were not mesh-checked.
- The FE models were not compared with a second FE code.
- No experiment has been done, so no result is validated against a physical actuator.

CHAPTER 6

CONCLUSIONS AND FUTURE WORK

6.1 Conclusions

1. The cross-section properties and the eccentric thrust moment of the actuator follow from statics; the moment $M = \pi P R^2 r^2 r_p/(R^2 - r^2)$ is uniform along the actuator for any material law.

2. The Euler–Bernoulli result, eq. (2.10), is the correct first-order model of a *hoop-reinforced* actuator: the COMSOL tip angle agrees with it within 1 % at $P/E = 0.017$ and 5 % at $P/E = 0.033$, and exceeds it at higher pressure.

3. For a *bare* silicone actuator the first-order bending vanishes by the factor $(1 - 2\nu)$. The bending that remains comes from second-order wall lengthening and is limited by ballooning. In COMSOL the bare actuator bends only 11 % of the beam value at $P/E = 0.017$, and its response steepens strongly with pressure.

4. Parametric, scripted COMSOL models with a follower cavity pressure, nearly incompressible neo-Hookean silicone and a weak-form hoop sleeve were set up for both actuators and can be re-run for the real design.

5. A tapered four-chamber actuator with a hoop sleeve bends smoothly under real pressure, to about 95° at $P/\mu = 0.64$ (15 kPa for $E = 0.07$ MPa); without the sleeve it balloons at $P/\mu = 0.50$. Deeper and wider chambers collapse; fillets reduce the bending slightly. Larger angles are not yet reliable.

6.2 Future work

1. Resolve the stress growth above $P/\mu \approx 0.64$ in the four-chamber model (local mesh refinement or hexahedral wall elements, then thicker side walls if it is physical).

2. Mesh-check the single-cavity models and continue the bare model to its ballooning limit (with a volume-controlled or arc-length solution).

3. Fix the real geometry and silicone; measure the material (uniaxial tests, Yeoh fit) and re-run the models.

4. Compute yaw and combined bending of the four-chamber actuator with real pressure.

5. Model contact with objects with a stiff contact formulation, then grasping.

6. Fabricate the actuator (moulding and casting [2]) and measure pressure, tip angle and cavity volume to validate the models.

REFERENCES

[1] B. Gorissen, D. Reynaerts, S. Konishi, K. Yoshida, J.-W. Kim and M. De Volder, "Elastic inflatable actuators for soft robotic applications," *Advanced Materials*, vol. 29, no. 43, p. 1604977, 2017. doi:10.1002/adma.201604977.

[2] A. D. Marchese, R. K. Katzschmann and D. Rus, "A recipe for soft fluidic elastomer robots," *Soft Robotics*, vol. 2, no. 1, pp. 7–25, 2015. doi:10.1089/soro.2014.0022.

[3] P. Polygerinos, Z. Wang, J. T. B. Overvelde, K. C. Galloway, R. J. Wood, K. Bertoldi and C. J. Walsh, "Modeling of soft fiber-reinforced bending actuators," *IEEE Transactions on Robotics*, vol. 31, no. 3, pp. 778–789, 2015. doi:10.1109/TRO.2015.2428504.

[4] F. Connolly, C. J. Walsh and K. Bertoldi, "Automatic design of fiber-reinforced soft actuators for trajectory matching," *Proceedings of the National Academy of Sciences*, vol. 114, no. 1, pp. 51–56, 2017. doi:10.1073/pnas.1615140114.

[5] J. M. Gere and B. J. Goodno, *Mechanics of Materials*, 8th ed. Stamford, CT: Cengage Learning, 2013.

[6] S. P. Timoshenko and J. N. Goodier, *Theory of Elasticity*, 3rd ed. New York: McGraw-Hill, 1970.

[7] L. R. G. Treloar, *The Physics of Rubber Elasticity*, 3rd ed. Oxford: Clarendon Press, 1975.

[8] G. A. Holzapfel, *Nonlinear Solid Mechanics: A Continuum Approach for Engineering*. Chichester: Wiley, 2000.

[9] O. H. Yeoh, "Some forms of the strain energy function for rubber," *Rubber Chemistry and Technology*, vol. 66, no. 5, pp. 754–771, 1993. doi:10.5254/1.3538343.

[10] R. J. Webster III and B. A. Jones, "Design and kinematic modeling of constant curvature continuum robots: a review," *International Journal of Robotics Research*, vol. 29, no. 13, pp. 1661–1683, 2010. doi:10.1177/0278364910368147.

[11] A. N. Gent, "Elastic instabilities in rubber," *International Journal of Non-Linear Mechanics*, vol. 40, nos. 2–3, pp. 165–175, 2005. doi:10.1016/j.ijnonlinmec.2004.05.006.

[12] COMSOL AB, *COMSOL Multiphysics Reference Manual*, version 6.4. Stockholm, Sweden.

[13] COMSOL AB, *Structural Mechanics Module User's Guide*, version 6.4. Stockholm, Sweden.

[14] Smooth-On, Inc., *Ecoflex Series Technical Bulletin* (Ecoflex 00-30: tensile strength 200 psi, 100 % modulus 10 psi, elongation at break 900 %).

APPENDIX A

SOFTWARE, DATA AND REPRODUCTION

Table A.1. Files behind the results of this report (paths relative to the project folder).

File	Content
`actuator_model.py`	beam model, Tiers 1–3
`comsol/bare/BuildActuatorBare.java`	bare single-cavity model; results in `comsol_bare_summary.csv` (Table 4.1)
`comsol/sleeved/BuildActuatorSleeve.java`	sleeved single-cavity model; results in `comsol_sleeve_summary.csv` (Table 4.1)
`comsol/tapered_4cavity_sleeved/BuildActuator4CavitySleeved.java`	four-chamber model (all cases)
`comsol/tapered_4cavity_sleeved/run_sleeved_study.py`	runs the four cases and writes `sleeved_study_summary.csv` and `sleeved_study_centreline.csv` (Table 4.2)
`report/btp_report/make_report_figures.py`	figures of Chapter 4 from the data files
`report/btp_report/render/RenderReport.java`	COMSOL images (Figures 3.2, 3.3, 4.2, 4.3, 4.6, 4.7) from the solved `.mph` files; also re-evaluates the single-cavity tip angles as a check
`report/btp_report/render/crop_renders.py`	crops those images for the report
`report/btp_report/render/Render180.java`, `crop_180.py`	images of the 20 kPa step (Figure 4.8) and its peak stresses

Commands (from the project folder):

```
comsolcompile BuildActuatorSleeve.java        (in comsol/sleeved/)
comsolbatch -inputfile BuildActuatorSleeve.class
python comsol/tapered_4cavity_sleeved/run_sleeved_study.py
python report/btp_report/make_report_figures.py
comsolcompile RenderReport.java               (in report/btp_report/render/)
comsolbatch -inputfile RenderReport.class
python render/crop_renders.py                 (in report/btp_report/)
```